SIT292 Module 3 Self-Assessment

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	Review
Question 1	4/4	
Question 2	1/1	
Question 3	1/1	
Question 4	1/1	
Question 5	1/1	
Question 6	1.5/2	
Total	9.5/10	(95%)

Congratulations! You have achieved the minimum threshold for this module's self-assessment.

Use the "Print this results summary" and save your attempt as a pdf. You will need the printout showing all questions for your module submission.

Performance Summary

Exam Name:	SIT292 Module 3 Self-Assessment
Session ID:	2350677760761549
Exam Start:	Tue Jul 29 2025 16:25:20
Exam Stop:	Tue Jul 29 2025 17:29:29
Time Spent:	1:04:09

Question 1

Starting with the matrix,

$$A = \begin{pmatrix} 2 & 2 & 3 \\ 4 & 3 & 1 \\ 1 & 1 & 2 \end{pmatrix}$$

The results following the first steps of the inversion algorithm are as follows:

Augmented matrix

$$\begin{pmatrix} 2 & 2 & 3 & 1 & 0 & 0 \\ 4 & 3 & 1 & 0 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{pmatrix}$$

Step 1.

$$\begin{pmatrix} 1 & 1 & 2 & 0 & 0 & 1 \\ 4 & 3 & 1 & 0 & 1 & 0 \\ 2 & 2 & 3 & 1 & 0 & 0 \end{pmatrix}$$

Step 2.

$$\begin{pmatrix} 1 & 1 & 2 & 0 & 0 & 1 \\ 0 & -1 & -7 & 0 & 1 & -4 \\ 0 & 0 & -1 & 1 & 0 & -2 \end{pmatrix}$$

Write down the resulting augmented matrix after performing the next step (step 3).

Rows: 3 Columns: 6

$$\begin{pmatrix} 1 & 0 & -5 & 0 & 1 & -3 \\ 0 & 1 & 7 & 0 & -1 & 4 \\ 0 & 0 & -1 & 1 & 0 & -2 \end{pmatrix}$$ ✔

Expected answer: $$\begin{pmatrix} 1 & 0 & -5 & 0 & 1 & -3 \\ 0 & 1 & 7 & 0 & -1 & 4 \\ 0 & 0 & -1 & 1 & 0 & -2 \end{pmatrix}$$

(Note that the step by step version of this problem is available in sub-module 3.2. At Step 3, you should not perform any additional operations to obtain a leading 1 in row 3 -- that would be part of Step 4)

Based on this and the Step 4 that would follow, the inverse matrix is:

$A^{-1} =$

Rows: 3 Columns: 3

$$\begin{pmatrix} -5 & 1 & 7 \\ 7 & -1 & -10 \\ -1 & 0 & 2 \end{pmatrix}$$

Expected answer: $\begin{pmatrix} -5 & 1 & 7 \\ 7 & -1 & -10 \\ -1 & 0 & 2 \end{pmatrix}$

Score: 4/4

Gap 0

Your answer is correct. You were awarded **2** marks.

Gap 1

Your answer is correct. You were awarded **2** marks.

You scored **4** marks for this part.

Question 2

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -6 \\ 0 & 0 & 1 \end{pmatrix}$$

Row operation applied

○ I. Interchange two rows

○ II. Multiply one row by a nonzero number

◉ III. Add a multiple of one row to a different row

✔

Expected answer:

○ I. Interchange two rows ○ II. Multiply one row by a nonzero number

◉ III. Add a multiple of one row to a different row

Inverse matrix

Rows: 3 Columns: 3

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{pmatrix}$$

✔

Expected answer: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \end{pmatrix}$

Score: 1/1 ✔

Gap 0

✔ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

✔ Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 3

For the following elementary matrix, indicate the corresponding elementary row operation and write the inverse.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{7} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Row operation applied

- ○ I. Interchange two rows
- ◉ II. Multiply one row by a nonzero number
- ○ III. Add a multiple of one row to a different row

✔

Expected answer:

- ○ I. Interchange two rows
- ◉ II. Multiply one row by a nonzero number
- ○ III. Add a multiple of one row to a different row

Inverse matrix

Rows: 3 Columns: 3

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$ ✔

Expected answer: $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Score: 1/1 ✔

Gap 0

✔ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

✔ Your answer is correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

Question 4

Find a matrix, E, such that $B = EA$ if

$$A = \begin{pmatrix} -5 & 9 \\ -7 & -6 \end{pmatrix}, B = \begin{pmatrix} -26 & -9 \\ -7 & -6 \end{pmatrix}$$

$E =$

Rows: 2 Columns: 2

$$\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

✔ Expected answer: $\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$

Score: 1/1 ✔

✔ Your answer is correct. You were awarded **1** mark.

You scored **1** mark for this part.

Question 5

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation.

Find $T\begin{pmatrix} -2 \\ -8 \\ -6 \end{pmatrix}$ if $T\begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 10 \\ -20 \end{pmatrix}$ and $T\begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix} = \begin{pmatrix} 3 \\ -14 \end{pmatrix}$.

Enter your result below.

$T\begin{pmatrix} -2 \\ -8 \\ -6 \end{pmatrix} =$

Rows: 2 Columns: 1

$$\begin{pmatrix} -14 \\ 12 \end{pmatrix}$$

✔ Expected answer: $\begin{pmatrix} -14 \\ 12 \end{pmatrix}$

Score: 1/1 ✔

✔ Your answer is correct. You were awarded **1** mark.

You scored **1** mark for this part.

Question 6

Consider the linear transformation $T:\mathbb{R}^2 \to \mathbb{R}^2$ and the effect of multiplying $T\begin{pmatrix} x \\ y \end{pmatrix}$.

a)

If

$$T = \begin{pmatrix} -\frac{4}{5} & -\frac{3}{5} \\ \frac{-6}{10} & \frac{4}{5} \end{pmatrix}$$

This linear mapping is

- ○ a rotation about the origin by θ
- ◉ a reflection through the line $y = mx$

✔

Expected answer:

- ○ a rotation about the origin by θ
- ◉ a reflection through the line $y = mx$

Give the value of θ if it is a rotation or the value of m if it is a reflection

-3 -3 ✔ Expected answer: -3 -3

*Give values of θ in terms of 'pi', e.g., pi/2, 3*pi/5 etc and express over the domain (-pi,pi].

**If your value involves surds, use 'sqrt', e.g. sqrt(2) for the square root of 2.

Score: 1/1 ✔

▼ Hide feedback

Gap 0

✔ You chose a correct answer. You were awarded **0.5** marks.

Gap 1

✔ Your answer is numerically correct. You were awarded **0.5** marks.

You scored **1** mark for this part.

b)

Identify a matrix that applies a rotation of pi/4 about the origin followed by a reflection through the line y = mx with m = 2

$\begin{pmatrix} 0.141 & 0.989 \\ 0.989 & -0.141 \end{pmatrix}$ ✔ Expected answer: $\begin{pmatrix} 0.141 & 0.990 \\ 0.990 & -0.141 \end{pmatrix}$

*Enter the entries of the matrix correct to 3 decimal places.

Score: 0.5/1 ✔

▼ Hide feedback

✔ One or more of the cells in your answer is incorrect, but you have been awarded marks for the rest. You were awarded **0.5** marks.

You scored **0.5** marks for this part.